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Enantioselective Dearomative Cyclization of Homotryptamines with Allenamides into Indolo[2,3-*b*]quinolines

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Abstract: We have developed a catalytic asymmetric dearomative cyclization of homotryptamine derivatives, furnishing enantioenriched indolo[2,3-*b*]quinoline scaffolds. The activation of allenamides in the presence of chiral phosphoric acids triggered the dearomative cyclization in up to 99% ee.

Keywords: dearomative cyclization; homotryptamine; indolo[2,3-*b*]quinolines; chiral Brønsted acid catalysis

Dearomatization of indoles through electrophilic activation has led to a range of powerful synthetic methods for polycyclic indolyl compounds.[1] In particular, there have been many elegant asymmetric dearomative cyclizations of tryptamine derivatives, affording pyrroloindolines under metal-[2] or organocatalytic conditions,[3,4] triggered by various electrophiles (Scheme 1A). Despite these remarkable developments on pyrroloindolines, cyclization of homotryptamine derivatives into indolo[2,3-*b*]quinolines have been much less studied (Scheme 1B).[5] Considering a variety of naturally occurring indole alkaloids share this core (Scheme 1C),[6] a highly enantioselective access to this scaffold is desired, but only a few methods are available to date.[7]

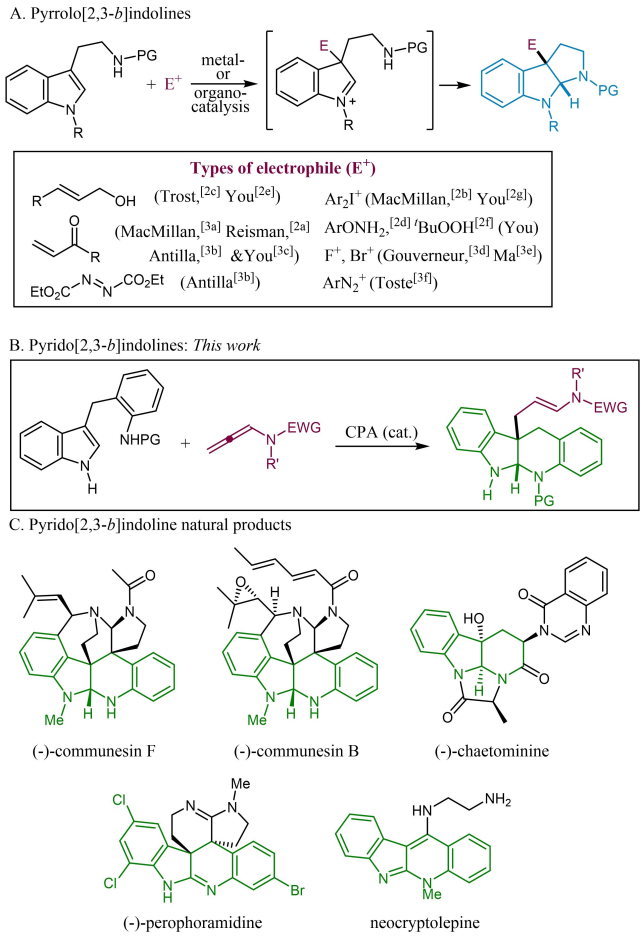
Inspired by the pioneering studies on the electrophilic activation of allenamides in gold-[8] and Brønsted acid-catalysis,[9] we initially posited that allenamides[10] would be good electrophilic sources triggering the dearomative cyclization of tryptamines as in Scheme 1A. To the best of our knowledge, only a single example of asymmetric dearomative cyclization of tryptamine derivatives with allenamides has been reported, employing chiral gold complexes.[8f]

Prompted by the lack of non-noble metal methods, we initially set out to search for conditions effecting the dearomative cyclization of tryptamine derivatives **1** with an oxazolidinone-derived allenamide **2a** (Scheme 2), employing chiral phosphoric acids (CPAs) (Figure 1). Use of 4 Å MS in toluene gave consistently higher yield and selectivity than other desiccants-solvent combinations.[11] Among various chiral phosphoric acids screened, (*R*)-**PA2** with 9-phenanthrenyl group at 3,3'-positions of the binol was identified as the most selective catalyst. These optimized conditions could be applied to various tryptamines, having substitution at C5-position (**1a–e**), affording the corresponding pyrroindolines **3a–e** in good enantioselectivity, although 6-Cl-substituted **1f** led to poor % ee.

However, a major problem with the cyclization of tryptamines **1** was the poor yield for **3**. Typically, the products were accompanied by unidentified byproducts. For example, with tryptamine **1g** (PG = *p*-Ns), significant amount of di-substitution product **3g'** (20%, 69% ee) formed, along with **3g** (49%, 70% ee). The formation of **3g'** is due to the nucleophilicity of *N*-atom of the indoline **3**, which consumes the iminium ion formed from the allenamide **2**.[2e,3b,8d] Unfortunately, simple change of the stoichiometry of **1** vs. **2** failed to improve the yield of **3** or **3'**.

Given this, we posited that the dearomative cyclization of homotryptamine derivatives would suffer less this problem, due to the increased steric crowding around the indolinyl *N*-atom, adjacent to the 5/6-fused bridgehead. Thus we set out to screen chiral phosphoric acid scaffolds (Table 1) and found (*S*)-**PA6** (TCyP) as the best catalyst (entry 6), furnishing indolo[2,3-*b*]quinoline **5a** in 77% ee at rt. Lowering the temperature to 0 °C further increased the enantioselectivity to 85% ee (entry 11). However, change of *N*-protecting groups as in **6** and **8** (entries 13–14), as well as employment of different allenamides **2b–e**[11] in place of **2a** were largely unsuccessful, affording lower yields and/or enantioselectivities. The product **9**

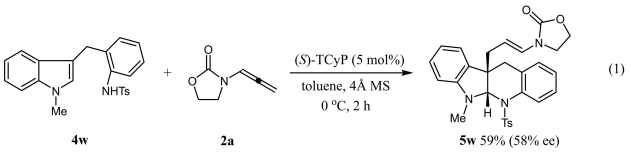
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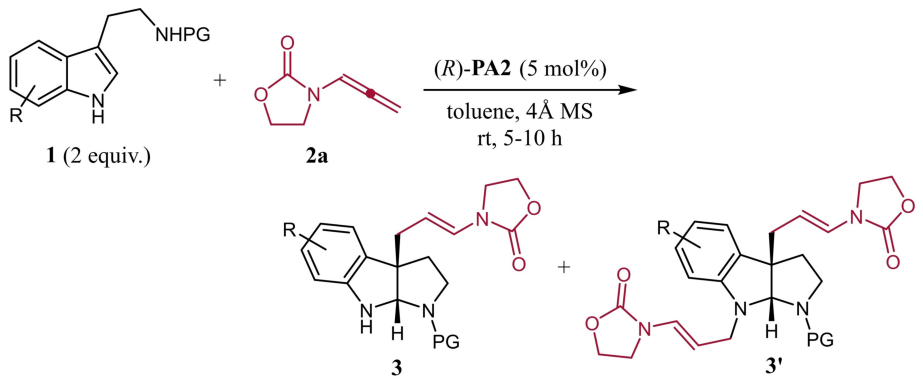
Scheme 1. Enantioselective dearomative cyclizations for polycyclic indolyl compounds.

having *N*-methoxycarbonyl (Moc) group, was obtained as an acyclic adduct.[11]

Based on these optimized conditions (entry 11, Table 1), various indolo[2,3-*b*]quinolines could be efficiently synthesized via dearomative cyclization of homotrypatmines **4** (Table 2). Synthesis of **5a** could be conducted in 1 mmol scale with essentially identical % ee and yield, confirming the practicality of the process. Substitution at the indole core (R^1) in **4** was well-tolerated at various positions. For example, a wide range of substituents with varying electron-demand were accommodated at C5 (**5b–g**). In addition, substitutions at C4 (**5h–i**) and at C7 (**5j**) positions were also satisfactory, but 6-Cl (**5k**) and 6-Me (**5l**) derivatives were obtained with markedly lower enantioselectivity. Interestingly, the reaction with **4g** gave di-substituted **5g'** exclusively, irrespective of the stoichiometry of **4g** and **2a**. Substitution at the aniline moiety (R^2) was also well-tolerated. For example, both electron-donating and withdrawing groups at C4 (**5k–o**) and C5 (**5p–r**), as well as at C3 (**5s**) and C6 (**5t**) were accommodated, affording good % ee and yields of **5**. The absolute configuration of the products was assigned based on the X-ray crystallographic structure of **5r** (Figure S1).[12]



In terms of the mechanism, several factors seemed to affect the enantio-determining step. A *N*-Me homotryptamine derivative **4w** gave a significantly lower % ee, suggesting that the indole N–H in **4** take part in the



1a (R = H, PG = Ts)
1b (R = 5-Me, PG = Ts)
1c (R = 5-OMe, PG = Ts)
1d (R = 5-Cl, PG = Ts)
1e (R = 5-F, PG = Ts)
1f (R = 6-Cl, PG = Ts)
1g (R = H, PG = *p*-Ns)

3a (42%, 90% ee)
3b (39%, 93% ee)
3c (40%, 88% ee)
3d (31%, 98% ee)
3e (36%, 76% ee)
3f (38%, 48% ee)
3g (49%, 70% ee) + **3g'** (20%, 69% ee)

Scheme 2. Enantioselective dearomative cyclization of tryptamine derivatives.

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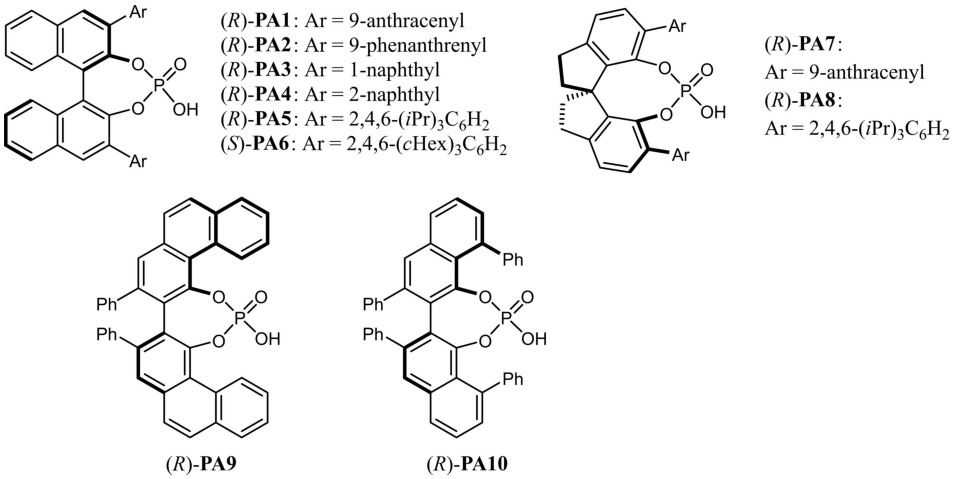
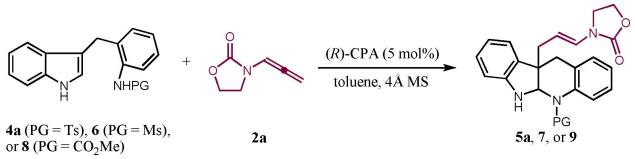


Figure 1. Chiral phosphoric acids used.

Table 1. Survey of reaction conditions for dearomative cyclization of homotryptamine derivatives.[a]



4a (PG = Ts), **6** (PG = Ms), or **8** (PG = CO_2Me) — **2a** — **5a**, **7**, or **9**

entry	**4** (PG)	CPA	cond.	yield	% ee
1	**4a** (Ts)	**PA1**	rt, 3 h	**5a**, 56	11
2	**4a** (Ts)	**PA2**	rt, 3 h	**5a**, 52	21
3	**4a** (Ts)	**PA3**	rt, 3.5 h	**5a**, 55	51
4	**4a** (Ts)	**PA4**	rt, 3.5 h	**5a**, 56	58
5	**4a** (Ts)	**PA5**	rt, 3.5 h	**5a**, 50	35
6	**4a** (Ts)	**PA6**[b]	rt, 3 h	**5a**, 62	77
7	**4a** (Ts)	**PA7**	rt, 3.5 h	**5a**, 57	60
8	**4a** (Ts)	**PA8**	rt, 3.5 h	**5a**, 58	56
9	**4a** (Ts)	**PA9**	rt, 4 h	**5a**, 60	17
10	**4a** (Ts)	**PA10**	rt, 4 h	**5a**, 55	5
11	**4a** (Ts)	**PA6**[b]	0 °C, 3 h	**5a**, 68	85
12	**4a** (Ts)	**PA6**[b]	-10 °C, 6 h	**5a**, 49	86
13	**6** (Ms)	**PA6**[b]	0 °C, 4 h	**7**, 51	67
14	**8**(Moc)	**PA6**[b]	0 °C, 6 h	**9**,[c] 58	49

[a] Homotryptamine **4** (0.1 mmol) and allenamide **2a** (0.05 mmol) was treated with **PA1**–**PA10** (5 mol%) in toluene (0.5 mL).
[b] (*S*)-**PA6** was used.

hydrogen bonding organization of the transition state. The enantioselectivity also depended on the *N*-protecting groups for aniline (**6** and **8**), underscoring the importance of hydrogen bonding of aniline N–H as well. Based on this, we propose the stereochemical model as follows (Figure 2). Dual activation of both allenamides **2** and tryptamine **4** is achieved by the hydrogen bonding organization depicted in Figure 2. Basic site of the phosphate in (*S*)-TCyP activates

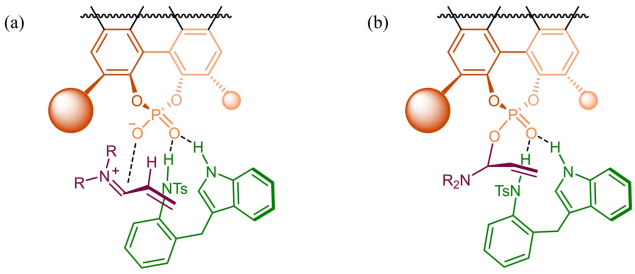


Figure 2. Possible stereochemical model for enantioinduction with (*S*)-TCyP.

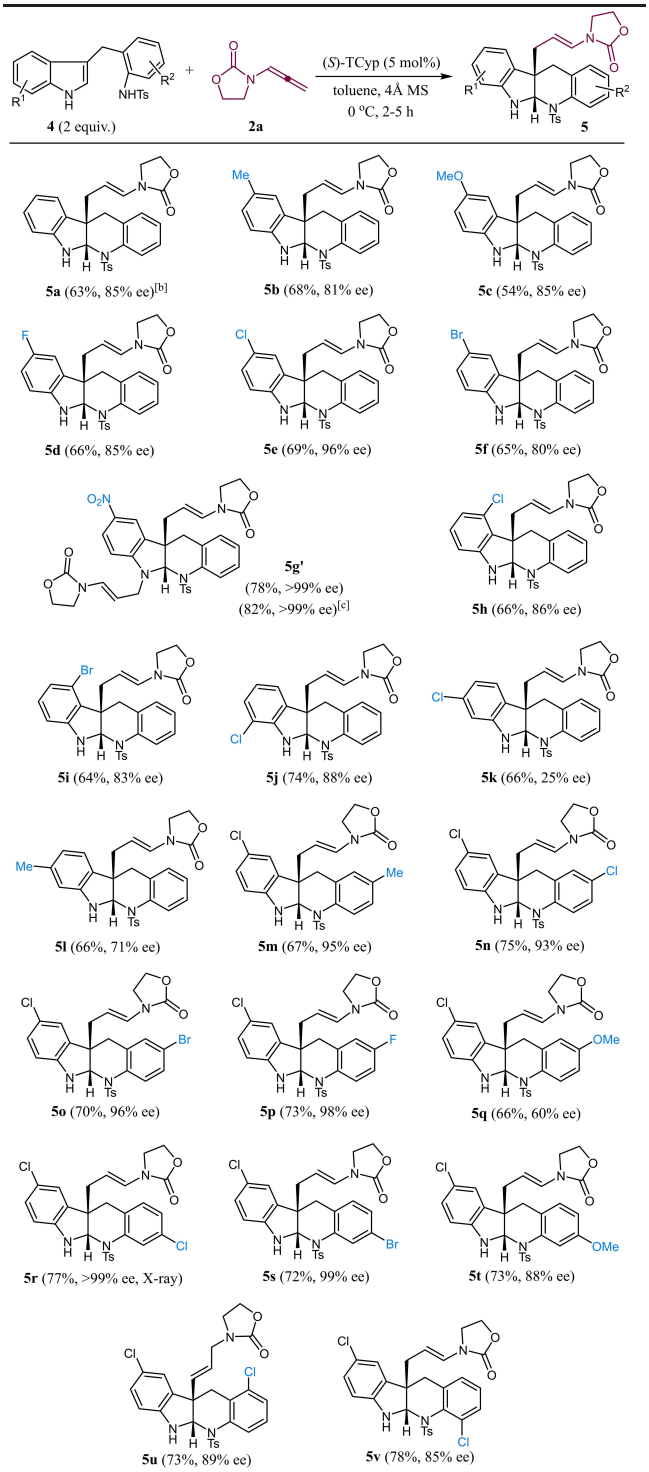
homotryptamines via dual hydrogen bonding with the indole and the aniline in **4**, whereas activation of allenamides **2** by protonation at C2 would form the α, β-unsaturated iminium ion, setting the stage for conjugate addition by the indole moiety of **4** (ionic model (a)). Alternatively, basic phosphate may form a covalent adduct with the iminium to form allylphosphate aminal (covalent model (b)), followed by the S_N2' attack of indole C3 carbon. In fact, Bandini and coworkers revealed that such possibility was more favourable than path (a), according to DFT computational study.[13]

Finally, the indolo[2,3-*b*]quinoline **5a** could be functionalized into alcohols **10** and **11** as in Scheme 3. Protection of aminal nitrogen, followed by hydrolysis afforded unstable aldehyde, which was reduced in-situ into **10** in excellent overall yields (84%) with no racemization. Oxidative cleavage of enamide in **5a**, followed by reduction furnished a homologous alcohol **11**. In addition, a regioselective fluorination of enamide unit,[14] followed by a reductive workup afforded 1,2-fluoroalcohol **12** as a diastereomeric mixture.

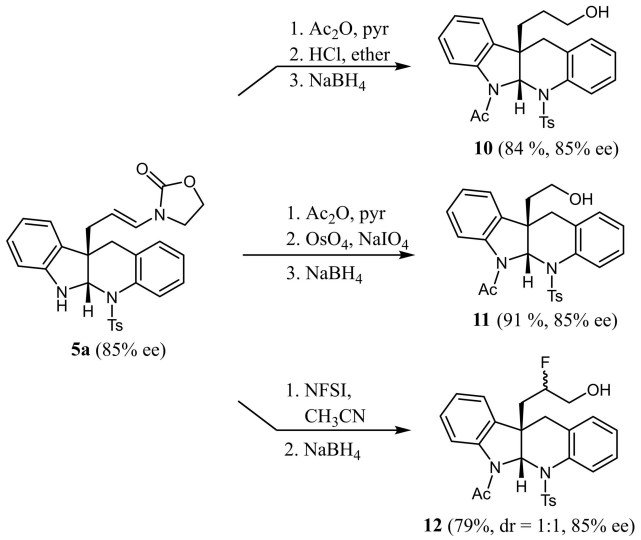
In summary, we have developed an organocatalytic route to enantio-enriched indolo[2,3-*b*]quinolines.

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Table 2. Enantioselective dearomative cyclization of homotryptamines for synthesis of indolo[2,3-*b*]quinolines.[a]



[a] Homotryptamine **4** (0.2 mmol) and allenamide **2a** (0.1 mmol) in toluene (1 mL).
[b] 1 mmol scale.
[c] With **4** (0.1 mmol) and **2a** (0.3 mmol).



Scheme 3. Transformation of products.

Lack of enantioselective cyclization of homotryptamines were addressed by use of allenamides as suitable electrophilic precursors. To the best of our knowledge, this protocol is the first general organocatalytic asymmetric cyclization of homotryptamine derivatives.

Experimental Section

To a flame-dried test tube was added 4 Å molecular sieve (20.0 mg), 3-(2λ^5-propa-1,2-dien-1-yl)oxazolidin-2-one **2a** (12.5 mg, 0.10 mmol), and *N*-(2-((1*H*-indol-3-yl)methyl) phenyl)-4-methylbenzenesulfonamide **4a** (75.3 mg, 0.20 mmol) in toluene (1.0 mL) and the mixture was kept stirred in an ice-bath. (*S*)-TCyP (**PA6**, 5.0 mg, 0.005 mmol) was then added and the mixture was stirred at 0 °C. After completion of the reaction (monitored by TLC), the mixture was directly loaded onto a silica gel column and flash column chromatography (EtOAc: Hex = 1:6 to 1:2) afforded 3-((*E*)-3-((5a*S*,10b*S*)-5-tosyl-5,5a,6,11-tetrahydro-10b*H*-indolo[2,3-*b*]quinolin-10b-yl)prop-1-en-1-yl)oxazolidin-2-one **5a** (34.1 mg, 68%) as a white solid (mp = 158–160 °C). The enantioselectivity was determined by Chiralpak® AD–H column using IPA:*n*Hex (30:70) as an eluent (1 mL/min).

Acknowledgements

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[11] See Supporting Information for details.

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